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Top Skills

Weights and Biases

LangChain

Fine Tuning

Certifications

Machine Learning Specialisation

SQL for Data Analysis: Advanced
SQL Querying Techniques

Python for Data Science and
Machine Learning Bootcamp

ChatGPT Prompt Engineering for
Developers

Amazon Web Services Cloud
Practitioner

Publications

Climate-Induced Physiological
Stress Drives Rainforest Mammal
Population Declines

Relationships between abiotic
factors, foliage chemistry and
herbivory in a tropical montane
ecosystem

Mountains magnify mechanisms in
climate change biology

Climate change threatens the future
of rain forest ringtail possums by
2050

The effect of forest gap dynamics on
tropical rainforest birds

Alejandro de la Fuente

AI Engineer | Data Scientist | Quantitative Ecology
Australia

Summary

I'm a scientist who builds intelligent systems, with rigorous foundations and measurable outcomes. My career began in quantitative ecology, spending 6+ years designing Bayesian forecasting and decision-support systems for complex, high-stakes problems. That work trained a very specific kind of thinking: how to reason under uncertainty, how to interrogate a model's assumptions before trusting its outputs, and how to design evaluations that expose failure before production does. This is rare in industry and increasingly valuable. Modern AI systems, from LLM-powered agents to ML pipelines, are fundamentally probabilistic. They demand the same rigour I spent years developing in research: structured evaluation, uncertainty quantification, and the analytical discipline to distinguish genuine performance from noise. Today I build end-to-end AI and data science systems, from model development through orchestration to deployment. I think at the system level, designing pipelines where components work together, degrade gracefully, and deliver against a measurable objective. I operate at the intersection of scientific modelling and production engineering, and that is where I am at my strongest.

Experience

James Cook University

Postdoctoral Fellow

October 2024 - Present (1 year 7 months)

Cairns, Queensland, Australia

I lead a postdoctoral research project on endangered spectacled flying foxes, combining field monitoring and predictive modelling to understand and forecast heat-driven mortality risk. The goal is to translate ecological and physiological evidence into a practical decision-support tool that helps conservation managers identify vulnerable roosts and prioritize interventions under increasing extreme heat.

University of Melbourne

Research Assistant

April 2024 - September 2024 (6 months)

Collected and curated museum specimen records (sampling, digitisation, metadata standardisation) and maintained the underlying database for quality-controlled downstream analysis.

Conducted initial spatiotemporal analyses to characterise patterns in occurrence/collection effort and prepare datasets for modelling and reporting.

James Cook University

Research Assistant

June 2019 - December 2019 (7 months)

Townsville, Queensland, Australia

Built reproducible workflows to estimate rainforest bird population trends from monitoring data, separating detection from true abundance using hierarchical/spatiotemporal models.

Linked abundance and trend estimates to climate and landscape predictors to produce decision-ready outputs for conservation prioritisation and reporting.

Education

James Cook University

Doctor of Philosophy - PhD, Ecology, Evolution, Systematics, and Population Biology · (August 2020 - April 2024)

University of Salamanca (Universidad de Salamanca)

Master of Science - MS, Ecology, Evolution, Systematics, and Population Biology · (September 2015 - June 2016)

University of Salamanca (Universidad de Salamanca)

Bachelor of Science - BS, Biology/Biological Sciences, General · (September 2010 - June 2014)